

1.

A p-n silicon step junction is doped with $N_A = 10^{19} \text{ cm}^{-3}$ and $N_D = 5 \times 10^{15} \text{ cm}^{-3}$.

- What is the built-in potential at room temperature?
- What is the built-in potential at 800K?
- What is the width of the depletion region at the n-side?
- What is the total depletion region width of the P-N junction?
- What is the maximum electric field in the depletion region?
- If a reverse bias of 4V is applied to the diode, what is the new maximum electric field in the junction?
- Calculate the percentage increase in the depletion width on the n-side and the p-side under 4V reverse bias.

(Use $n_i = 10^{10} \text{ cm}^{-3}$ at 300K)

[0.877 V, 0.209 V, 47.26 μm , $3.614 \times 10^4 \text{ V/cm}$, $8.627 \times 10^4 \text{ V/cm}$, 139%, 139%]

$$N_A = 10^{19} \text{ cm}^{-3}, N_D = 5 \times 10^{15} \text{ cm}^{-3}$$

a) room temperature $T_0 = 300 \text{ K} \Rightarrow n_i = 10^{10} \text{ cm}^{-3}$

$$V_{bi} = \frac{kT}{q} \cdot \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

$$= \frac{(1.38 \times 10^{-23} \text{ J/K}) (300 \text{ K})}{1.6 \times 10^{-19} \text{ C}} \cdot \ln \left(\frac{10^{19} \times 5 \times 10^{15}}{10^{20}} \right)$$

$$= 0.877 \text{ V}$$

b) $T_1 = 800 \text{ K}$

$$n_i = 2 \left(\frac{2\pi kT}{h^2} \right)^{\frac{3}{2}} \cdot (m_n^* m_p^*)^{\frac{3}{4}} \cdot \exp \left(-\frac{E_g}{2kT} \right)$$

$$n_i(T_0) = A \cdot (T_0)^{\frac{3}{2}} \cdot \exp \left(-\frac{E_g}{2k} \cdot \frac{1}{T_0} \right)$$

$$n_i(T_1) = A \cdot (T_1)^{\frac{3}{2}} \cdot \exp \left(-\frac{E_g}{2k} \cdot \frac{1}{T_1} \right)$$

$$\Rightarrow n_i^2(T_1) = n_i^2(T_0) \cdot \left(\frac{T_1}{T_0} \right)^3 \exp \left(-\frac{E_g}{k} \left(\frac{1}{T_0} - \frac{1}{T_1} \right) \right)$$

$$\Rightarrow n_i^2(800 \text{ K}) = 2.41 \times 10^{33}$$

$$V_{bi}(800 \text{ K}) = \frac{kT}{q} \cdot \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

$$= \frac{1.38 \times 10^{-23} \cdot 800}{1.6 \times 10^{-19}} \cdot \ln \left(\frac{5 \times 10^{34}}{n_i^2(800 \text{ K})} \right)$$

$$\approx 0.209 \text{ V}$$

$$c) W_n = \sqrt{\frac{2\epsilon_s \cdot N_A}{q \cdot N_A N_D + N_D^2}} \cdot V_{bi}$$

$$= \sqrt{\frac{\epsilon_s \times 2}{1.6 \times 10^{-19}} \cdot \frac{10^{19} \times 10^{-6}}{5 \times 10^{34} + 2.5 \times 10^{31}}} \cdot (0.877V)$$

$$= 47.26 \mu m$$

$$d) W_D = W_n \cdot \frac{N_A + N_D}{N_A} \approx 47.26 \mu m$$

$$e) E_{max} = \frac{q \cdot N_D \cdot x_n}{\epsilon_s} \approx 3.614 \times 10^4 V/cm$$

$$f) E_{max}(V_R) = E_{max}(0) \sqrt{\frac{V_{bi} + V_R}{V_{bi}}}$$

$$E_{max}(4V) \approx 8.627 \times 10^4 V/cm$$

$$g) W(V) = W(0) \cdot \sqrt{\frac{V_{bi} + V}{V_{bi}}}$$

$$\eta = \left(\sqrt{\frac{V_{bi} + V_R}{V_{bi}}} - 1 \right) \times 100\%$$

$$\Rightarrow \eta = 139\% \quad \rightarrow \quad n\text{-side: } 139\%, \quad p\text{-side: } 139\%$$

2.

An abrupt silicon pn junction at zero bias has dopant concentrations of $N_D = 10^{17} \text{ cm}^{-3}$ and $N_A = 5 \times 10^{15} \text{ cm}^{-3}$. $T = 300 \text{ K}$. (a) Calculate the Fermi level on each side of the junction with respect to the intrinsic Fermi level. (b) Sketch the equilibrium energy-band diagram for the junction and determine V_{bi} from the diagram and the results of part (a). (c) Calculate V_{bi} using Equation (7.10), and compare the results to part (b). (d) Determine x_n , x_p , and the peak electric field for this junction.

(Use $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\epsilon = 11.7$ for silicon at 300K)

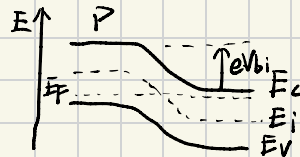
[0.3294 eV, 0.407 eV, 0.7364 V, 0.7363 V, 0.426 μm , 0.0213 μm , $3.29 \times 10^4 \text{ V/cm}$]

$$N_A = 10^{17} \text{ cm}^{-3}, N_D = 5 \times 10^{15} \text{ cm}^{-3}, T = 300K$$

$$a) \text{ n-type: } E_{Fn} - E_i = kT \cdot \ln\left(\frac{N_D}{n_i}\right) \approx 0.407 \text{ eV} \quad \text{above}$$

$$\text{p-type: } E_i - E_{Fp} = kT \cdot \ln\left(\frac{N_A}{n_i}\right) \approx 0.3294 \text{ eV} \quad \text{below}$$

$$b) V_{bi} = \frac{E_{Fn} - E_{Fp}}{q} = 0.7364 V$$



$$c) V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

$$\Rightarrow V_{bi} = 0.0259 \cdot \ln \left(\frac{5 \times 10^{22}}{(1.5 \times 10^{10})^2} \right) \approx 0.736 \text{ V}$$

Compare : nearly equal

$$d) \epsilon_s = \epsilon_r \cdot \epsilon_0 = 1.036 \times 10^{-12} \text{ F/cm}$$

$$\Rightarrow x_n = \sqrt{\frac{2\epsilon_s}{q} \frac{N_A}{N_A N_D + N_D^2} V_{bi}} \approx 0.0213 \mu\text{m}$$

$$\Rightarrow x_p = \sqrt{\frac{2\epsilon_s}{q} \frac{N_D}{N_A N_D + N_A^2} V_{bi}} \approx 0.426 \mu\text{m}$$

$$\Rightarrow W = x_n + x_p = 0.447 \mu\text{m}$$

$$E_{\max} = \frac{q N_D \cdot x_n}{\epsilon_s} = 3.29 \times 10^4 \text{ V/cm}$$

3.

3.

(a) Consider a uniformly doped silicon pn junction at $T = 300 \text{ K}$. At zero bias, 25 percent of the total space charge region is in the n-region. The built-in potential barrier is $V_{bi} = 0.710 \text{ V}$. Determine (i) N_A , (ii) N_D , (iii) x_n , (iv) x_p , and (v) $|E_{\max}|$. (b) Repeat part (a) for a GaAs pn junction with $V_{bi} = 1.180 \text{ V}$.

(Use $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\epsilon = 11.7$ for silicon and $n_i = 1.8 \times 10^6 \text{ cm}^{-3}$ and $\epsilon = 13.1$ for GaAs)

$[7.766 \times 10^{15} \text{ cm}^{-3}, 2.33 \times 10^{16} \text{ cm}^{-3}, 0.0993 \mu\text{m}, 0.2979 \mu\text{m}, 3.58 \times 10^4 \text{ V/cm},$

a) silicon PN Junction $x_n = 0.25W$, $x_p = 0.75W$

$$\frac{N_A}{N_D} = \frac{x_n}{x_p} = \frac{0.25}{0.75} = \frac{1}{3} \quad , \quad N_D = 3N_A$$

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2(\text{Si})} \right) = 0.710 \quad \checkmark$$

$$i) \Rightarrow N_A^2 = 6.03 \times 10^{31} \Rightarrow N_A = 7.766 \times 10^{15} \text{ cm}^{-3}$$

$$ii) \Rightarrow N_D = 3N_A \Rightarrow N_D = 2.33 \times 10^{16} \text{ cm}^{-3}$$

$$\epsilon_{Si} = \epsilon_r \cdot \epsilon_0 = 1.036 \times 10^{-12} \text{ F/cm}$$

$$W = \sqrt{\frac{2\epsilon_{Si}}{q} \frac{N_A + N_D}{N_A N_D} V_{bi}} = 0.397 \mu\text{m}$$

$$\text{iii)} \quad x_n = 0.25 W = 0.0993 \mu\text{m}$$

$$\text{iv)} \quad x_p = 0.75 W = 0.2979 \mu\text{m}$$

$$\text{v)} \quad |E_{\max}| = \frac{2V_{bi}}{W} = 3.58 \times 10^4 \text{ V/cm}$$

$$\text{b)} \quad \text{GaAs} \quad n_i(\text{GaAs}) = 1.8 \times 10^6 \text{ cm}^{-3}$$

$$V_{bi} = \frac{kT}{q} \cdot \ln \left(\frac{3N_A^2}{n_i^2} \right)$$

$$\text{i)} \quad N_A = 8.127 \times 10^{15} \text{ cm}^{-3}$$

$$\text{ii)} \quad N_D = 2.438 \times 10^{16} \text{ cm}^{-3}$$

$$W = \sqrt{\frac{2\varepsilon_{\text{GaAs}}}{q} \cdot \frac{N_A + N_D}{N_A \cdot N_D} \cdot V_{bi}}$$

$$= 0.5297 \mu\text{m}$$

$$\text{iii)} \quad x_n = 0.25 W = 0.1324 \mu\text{m}$$

$$\text{iv)} \quad x_p = 0.75 W = 0.3973 \mu\text{m}$$

$$\text{v)} \quad |E_{\max}| = \frac{2V_{bi}}{W} = 4.45 \times 10^4 \text{ V/cm}$$

4.

4. (a) The peak electric field in a reverse-biased silicon pn junction is $|E_{\max}| = 3 \times 10^5 \text{ V/cm}$. The doping concentrations are $N_A = 4 \times 10^{15} \text{ cm}^{-3}$ and $N_D = 4 \times 10^{17} \text{ cm}^{-3}$. Find the magnitude of the reverse-biased voltage. (b) Repeat part (a) for $N_A = 4 \times 10^{16} \text{ cm}^{-3}$ and $N_D = 4 \times 10^{17} \text{ cm}^{-3}$. (c) Repeat part (a) for $N_A = N_D = 4 \times 10^{17} \text{ cm}^{-3}$. (Use $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\varepsilon = 11.7$ for silicon)

[0.766 V, 72.8 V, 0.826 V, 7.18 V, 0.886 V, 0.57V]

$$\text{a)} \quad N_A = 4 \times 10^{15} \text{ cm}^{-3}, \quad N_D = 4 \times 10^{17} \text{ cm}^{-3}$$

$$|E_{\max}| = 3 \times 10^5 \text{ V/cm}$$

$$\Rightarrow V_{bi} = \frac{kT}{q} \cdot \ln \left(\frac{N_A \cdot N_D}{n_i^2} \right) = 0.766 \text{ V}$$

$$V_{\text{total}} = \frac{\varepsilon_s |E_{\max}|^2}{2q} \cdot \frac{N_A + N_D}{N_A \cdot N_D} = 73.6 \text{ V}$$

$$\Rightarrow V_R = V_{total} - V_{bi} = 72.8 V$$

$$b) N_A = 4 \times 10^{16} \text{ cm}^{-3}, N_D = 4 \times 10^{17} \text{ cm}^{-3}$$

$$\Rightarrow V_{bi} = \frac{kT}{q} \cdot \ln \left(\frac{N_A \cdot N_D}{n_i^2} \right) = 0.826 V$$

$$V_{total} = \frac{\epsilon \cdot E_{max}^2}{2q} \left(\frac{N_A + N_D}{N_A \cdot N_D} \right) = 8.01 V$$

$$\Rightarrow V_R = V_{total} - V_{bi} = 7.18 V$$

$$c) N_A = N_D = 4 \times 10^{17} \text{ cm}^{-3}$$

$$\Rightarrow V_{bi} = \frac{kT}{q} \cdot \ln \left(\frac{N_A \cdot N_D}{n_i^2} \right) = 0.886 V$$

$$V_{total} = \frac{\epsilon \cdot E_{max}^2}{2q} \cdot \left(\frac{N_A + N_D}{N_A \cdot N_D} \right) = 1.457 V$$

$$\Rightarrow V_R = V_{total} - V_{bi} = 0.57 V$$

5.

*A Si p-n junction has donor and acceptor doping concentrations N_A and N_D of 10^{16} cm^{-3} and 10^{18} cm^{-3} , respectively. Assume $kT = 0.026 \text{ eV}$ at 300 K, $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\epsilon_r = 11.8$. Calculate under thermal equilibrium conditions

- the Fermi energy levels and draw an equilibrium energy band diagram.
- the contact potential V_0 (compare it to that obtained in part (a))
- the depletion width W
- the penetration depth into the n-side and p-side
- peak electric field ξ_0 at the junction and
- the space charge in each side of the depletion region, given a cross-sectional area of 10^{-4} cm^2

Comment on depletion width and the penetration depth obtained in relation to the doping concentration on each side. (Ans: a) $E_{ip} - E_{fp} = 0.468 \text{ eV}$, $E_{fn} - E_{in} = 0.349 \text{ eV}$, b) 0.817 V , c) $0.328 \text{ } \mu\text{m}$, d) $x_{po} = 3.25 \times 10^{-3} \text{ } \mu\text{m}$, $x_{no} = 0.325 \text{ } \mu\text{m}$, e) $-5 \times 10^4 \text{ V/cm}$, f) $5.2 \times 10^{-12} \text{ C}$)

$$N_A = 10^{18} \text{ cm}^{-3}, N_D = 10^{16} \text{ cm}^{-3}, T = 300 K$$

$$\epsilon_r = 11.8 \Rightarrow \epsilon_s = 1.045 \times 10^{-12} \text{ F/cm}, q = 1.6 \times 10^{-19} \text{ C}, A = 10^{-4} \text{ cm}^2$$

$$a) \text{ p-area: } E_{ip} - E_{fp} = kT \cdot \ln \left(\frac{N_A}{n_i} \right) \approx 0.468 \text{ eV}$$

$$\text{ n-area: } E_{fn} - E_{in} = kT \cdot \ln \left(\frac{N_D}{n_i} \right) \approx 0.349 \text{ eV}$$

$$b) V_{bi} = \frac{E_{fn} - E_{fp}}{q} = 0.817 V$$

$$c) W = \sqrt{\frac{2\epsilon_s}{q} \cdot \frac{N_A + N_D}{N_A \cdot N_D} V_{bi}} = 0.328 \text{ } \mu\text{m}$$

$$d) N_A \cdot X_{po} = N_D \cdot X_{no} \Rightarrow \frac{X_{po}}{X_{no}} = \frac{N_D}{N_A} = \frac{10^{16}}{10^{18}} = 0.01$$

$$W = X_{po} + X_{no} = 101 X_{po}$$

$$X_{po} = \frac{W}{101} = 3.25 \text{ nm} = 3.25 \times 10^{-3} \text{ mm}$$

$$\Rightarrow X_{no} = 100 \cdot X_{po} = 0.325 \text{ mm}$$

$$e) E_{\max} = \frac{2 V_{bi}}{W} = 4.98 \times 10^4 \text{ V/cm}$$

$$\Rightarrow E_0 = -5 \times 10^4 \text{ V/cm}$$

$$f) |Q| = q \cdot N_A \cdot A \cdot X_{po} \approx 5.2 \times 10^{-12} \text{ C}$$

6.

6.

Calculate the forward bias V_F of an ideal p-n junction diode at 300 K to result in a current of 15 mA if the reverse saturation current I_0 is (a) 5 μA and (b) 8 pA. (0.2 V, 0.53 V)

$$I_F \approx I_0 \cdot \exp\left(\frac{q \cdot V_F}{kT}\right), I_F = 15 \text{ mA}$$

$$\hookrightarrow V_F = \frac{kT}{q} \cdot \ln\left(\frac{I_F}{I_0}\right)$$

$$a) I_0 = 5 \mu\text{A}$$

$$\frac{I_F}{I_0} = 3 \times 10^3 \Rightarrow V_F = 0.0259 \times \ln 3000 \approx 0.20 \text{ V}$$

$$b) I_0 = 8 \text{ pA}$$

$$\frac{I_F}{I_0} = \frac{15 \times 10^3}{8 \times 10^{-12}} = \frac{15}{8} \times 10^9$$

$$\Rightarrow V_F = 0.0259 \times \ln\left(\frac{15}{8} \times 10^9\right) \approx 0.53 \text{ V}$$

7.

*A Si p-n junction has donor and acceptor doping concentrations N_D and N_A of 10^{16} cm^{-3} and 10^{18} cm^{-3} , respectively. Assume $kT = 0.026 \text{ eV}$ at 300 K , $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\epsilon_r = 11.8$. Calculate the following parameters at (i) FB of 0.65 V and (ii) RB of 2 V .

- contact potentials under both FB and RB. Draw the band diagrams. Clearly indicate the extent of band bending.
- the depletion width W
- the penetration depth into the n-side and p-side
- peak electric field ξ_m at the junction and
- the space charge in each side of the depletion region, given a cross-sectional area of 10^{-3} cm^2

(Ans: a) $V_c = 0.167 \text{ V}$, $V_c = 2.817 \text{ V}$, b) $0.148 \mu\text{m}$, $0.610 \mu\text{m}$, c) $1.47 \times 10^{-3} \mu\text{m}$, $0.147 \mu\text{m}$, $6.04 \times 10^{-3} \mu\text{m}$, $0.604 \mu\text{m}$, d) $-2.251 \times 10^4 \text{ V/cm}$, $-9.25 \times 10^4 \text{ V/cm}$, e) 2.35 pC , 9.66 pC)

$$n_D = 10^{16} \text{ cm}^{-3}, N_A = 10^{18} \text{ cm}^{-3}, kT = 0.026 \text{ eV at } 300 \text{ K}, n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$\epsilon = \epsilon_r \cdot \epsilon_0 = 11.8 \cdot \epsilon_0 = 1.045 \times 10^{-12} \text{ F/cm}$$

$$V_F = 0.65 \text{ V}, V_R = 2 \text{ V}$$

$$a) V_{bi} = \frac{kT}{q} \cdot \ln\left(\frac{N_A N_D}{n_i^2}\right) = 0.817 \text{ V}$$

$$i) V_{eff} = V_{bi} - V_F = 0.167 \text{ V}$$

$$ii) V_{eff} = V_{bi} + V_R = 2.817 \text{ V}$$

$$b) C = \sqrt{\frac{2\epsilon}{q} \left(\frac{N_A + N_D}{N_A N_D} \right)} = 3.63 \times 10^{-5}$$

$$i) W_{FB} = C \cdot \sqrt{V_{eff}} = 0.148 \mu\text{m}$$

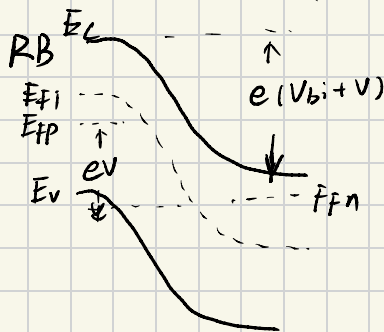
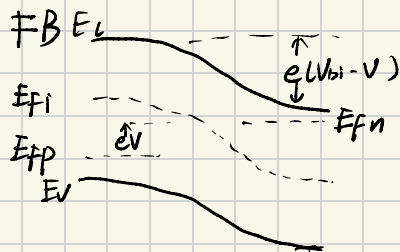
$$ii) W_{RB} = C \cdot \sqrt{V_{eff}} = 0.610 \mu\text{m}$$

$$c) \frac{N_A}{N_D} = 100 \Rightarrow x_p = 0.01 w, x_n = 0.99 w$$

$$i) x_{p,FB} = 1.47 \times 10^{-3} \mu\text{m}$$

$$x_{n,FB} = 0.147 \mu\text{m}$$

$$ii) x_{p,RB} = 6.04 \times 10^{-3} \mu\text{m}$$



$$x_n, R_B = 0.604 \mu\text{m}$$

$$d) i) E_{\max, FB} = \frac{2 \times 0.167}{0.148 \times 10^{-4}} = -2.25 \times 10^4 \text{ V/cm}$$

$$ii) E_{\max, RB} = \frac{2 \times 2.817}{0.610 \times 10^{-4}} = -9.25 \times 10^4 \text{ V/cm}$$

$$e) |Q| = q \cdot N_A \cdot A \cdot x_p \quad q \cdot N_A \cdot A = 1.6 \times 10^{-5}$$

$$i) |Q_{FB}| = 1.6 \times 10^{-5} \times 1.47 \times 10^{-7} = 2.35 \text{ pC}$$

$$ii) |Q_{RB}| = 1.6 \times 10^{-5} \times 6.04 \times 10^{-7} = 9.66 \text{ pC}$$

8.

Consider a silicon p-n junction with donor and acceptor doping concentrations N_d and N_a of 10^{15} cm^{-3} and 10^{18} cm^{-3} , respectively. A forward bias of 0.52 V is applied to the p-n junction. Calculate the minority-carrier concentrations at the depletion layer-edges. Assuming a very weak electric field in the neutral regions of the semiconductor (outside the depletion region), estimate the ratio of the majority carrier to minority carrier drift currents at the depletion layer edge and deep inside the n-region. Assume that the diode is at room temperature and $\mu_n = 1350 \text{ cm}^2/\text{V}\cdot\text{s}$, $\mu_p = 450 \text{ cm}^2/\text{V}\cdot\text{s}$ and $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$. (10^{14} cm^{-3} , 10^{11} cm^{-3} , 30, 1.34×10^{10})

$$N_A = 10^{18} \text{ cm}^{-3}, N_D = 10^{15} \text{ cm}^{-3}, V_F = 0.52 \text{ V}$$

$$a) p_{n0} = \frac{n_i^2}{N_D} = 2.25 \times 10^5 \text{ cm}^{-3}$$

$$n_{p0} = \frac{n_i^2}{N_A} = 2.25 \times 10^2 \text{ cm}^{-3}$$

$$\Rightarrow p_n(x_n) = p_{n0} \cdot \exp\left(\frac{V_F}{V_T}\right) \quad V_T = \frac{kT}{q}$$

$$\Rightarrow p_n(x_n) = 10^{14} \text{ cm}^{-3}$$

$$\Rightarrow n_p(-x_p) = n_{p0} \cdot \exp\left(\frac{V_F}{V_T}\right) = 10^{11} \text{ cm}^{-3}$$

$$b) \frac{J_{n, \text{drift}}}{J_{p, \text{drift}}} = \frac{q \cdot \mu_n \cdot N_D \cdot E}{q \cdot \mu_p \cdot p_n(x_n) \cdot E} = \frac{\mu_n \cdot N_D}{\mu_p \cdot p_n(x_n)} = 30$$

deep $\frac{J_{n, \text{drift}}}{J_{p, \text{drift}}} = \frac{\mu_n \cdot N_D}{\mu_p \cdot P_{no}} = 1.34 \times 10^{10}$

9.

*An abrupt Si p-n junction diode is fabricated with 1 Ω -cm p-type and 0.2 Ω -cm n-type materials in which the minority carrier lifetimes are $\tau_n = 10^{-6}$ sec and $\tau_p = 10^{-8}$ sec respectively. A forward bias of 0.55 V is applied across the junction.

- Calculate the donor concentration of the n-type material and the acceptor concentration of the p-type material
- Calculate the minority carrier diffusion lengths
- Calculate the density of minority carriers at both edges of the space charge region
- Sketch the majority and minority carrier distributions as functions of distance from the junction indicating the minority carrier diffusion lengths on your distance axis

- Sketch the majority and minority carrier currents outside the space charge region as functions of distance from the junction
- Calculate the location of the plane at which the majority carrier and minority carrier currents are equal in magnitude

Assume that $T = 300\text{K}$, $kT = 0.026 \text{ eV}$, and for Si, $\mu_n = 0.145 \text{ m}^2/\text{V-sec}$, $\mu_p = 0.045 \text{ m}^2/\text{V-sec}$ and $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$.

(Ans : a) $2.16 \times 10^{22} \text{ m}^{-3}$, $1.39 \times 10^{22} \text{ m}^{-3}$, b) $6.14 \times 10^{-5} \text{ m}$, $3.42 \times 10^{-6} \text{ m}$, c) $1.6 \times 10^{19} \text{ m}^{-3}$, $2.49 \times 10^{19} \text{ m}^{-3}$, f) $1.53 \times 10^{-6} \text{ m}$ from the edge at the n-side)

a) $P : N_A = \frac{1}{q \cdot \mu_p \cdot \rho_p} = 1.39 \times 10^{22} \text{ m}^{-3}$

$N : N_D = \frac{1}{q \cdot \mu_n \cdot \rho_n} = 2.16 \times 10^{22} \text{ m}^{-3}$

b) $D_n = \mu_n \cdot V_T = 37.6 \text{ cm}^2/\text{s}$

$D_p = \mu_p \cdot V_T = 11.7 \text{ cm}^2/\text{s}$

$\Rightarrow L_n = \sqrt{D_n \cdot \tau_n} = 6.14 \times 10^{-5} \text{ m}$

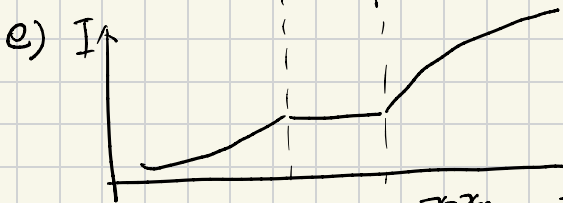
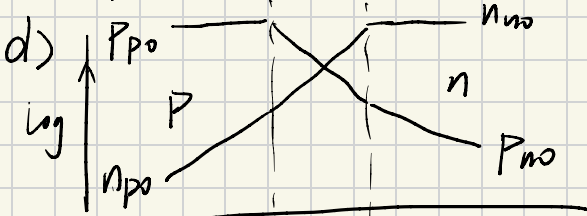
$\Rightarrow L_p = \sqrt{D_p \cdot \tau_p} = 3.42 \times 10^{-6} \text{ m}$

c) $P_{no} = \frac{n_i^2}{N_D} = 1.04 \times 10^4 \text{ cm}^{-3}$

$$n_{p0} = \frac{n_i^2}{N_A} = 1.62 \times 10^4 \text{ cm}^{-3}$$

$$\Rightarrow p_n(x_n) = p_{n0} \cdot \exp\left(\frac{V_F}{V_T}\right) = 1.6 \times 10^9 \text{ cm}^{-3}$$

$$\Rightarrow n_p(-x_p) = n_{p0} \cdot \exp\left(\frac{V_F}{V_T}\right) = 2.5 \times 10^9 \text{ cm}^{-3}$$



$$f) J_p(x) = J_p(x_n) \cdot e^{-\frac{x-x_n}{L_p}} = \frac{J}{2}$$

$$\Rightarrow \Delta x = 0.447 L_p \Rightarrow \Delta x = 1.53 \times 10^{-6} \text{ m from edge at n}$$

10

10.

A silicon p-n junction diode has the following parameters at 300 K

$$D_n = 25 \text{ cm}^2/\text{sec} \quad D_p = 10 \text{ cm}^2/\text{sec} \quad \tau_n = \tau_p = 0.5 \mu\text{s} \quad \epsilon_r = 11.7$$

$$n_i = 1.5 \times 10^{10} / \text{cm}^3$$

Given that the electron diffusion current is 20 A/cm² and the hole diffusion current is 5 A/cm² at a forward bias of 0.65 V, determine the doping densities of the p and the n regions of the diode. ($N_a = 10^{15} / \text{cm}^3$ and $N_d = 2.5 \times 10^{15} / \text{cm}^3$)

$$L_n = \sqrt{D_n \tau_n} = 3.54 \times 10^{-3} \text{ cm}, \quad L_p = \sqrt{D_p \tau_p} = 2.24 \times 10^{-3} \text{ cm}$$

$$p_{n0} = \frac{n_i^2}{N_D}, \quad n_{p0} = \frac{n_i^2}{N_A} \Rightarrow J_n = q \cdot \frac{D_n n_{p0}}{L_n} \cdot \exp\left(\frac{V_F}{V_T}\right), \quad J_p = q \cdot \frac{D_p p_{n0}}{L_p} \cdot \exp\left(\frac{V_F}{V_T}\right)$$

$$\Rightarrow \frac{J_n}{J_p} = \frac{D_n N_D \cdot L_p}{D_p \cdot N_A \cdot L_n} = 4$$

$$\Rightarrow \frac{N_D}{N_A} = 4 \cdot \frac{D_p L_n}{D_n \cdot L_p} = 2.53$$

$$\hookrightarrow N_D = q \cdot \frac{D_n \cdot n_{p0}}{L_n} \cdot \exp\left(\frac{V_F}{V_T}\right) = 2.5 \times 10^{15} \text{ cm}^{-3}$$

$$N_A = 1.0 \times 10^{15} \text{ cm}^{-3}$$